Multivariate Lab-8 **24-03-2026**

In the first phase of a study of the cost of transporting milk from farms to dairy plants, a survey was taken of firms engaged in milk transportations. Cost data on X_1 = fuel, X_2 = repair and X_3 = capital, all measured on a per mile basis, are presented in below table for n_1=15 gasoline and n_2=15 diesel trucks.

Table-1: Milk Transportation Cost Data

Gasoline Trucks			Diesel Trucks		
X_1	X_2	X_3	X_1	X_2	X_3
16	12	11	8	12	9
7	3	4	7	5	17
10	2	10	10	3	11
4	6	8	10	15	6
11	5	11	13	4	29
14	6	10	10	13	11
13	11	11	6	9	19
13	14	9	11	10	14
29	15	3	9	3	14
13	8	10	10	5	21
7	6	8	11	18	35
10	4	9	12	12	17
10	5	10	9	13	21
11	6	8	8	10	17
12	14	14	8	6	16

Test whether mean cost vectors for Gasoline Trucks and Diesel Trucks are same or not at $\alpha = 0.01$ level of significance [$F_{3,\,14}(0.01) = 5.56$].

Step 1 :- Hypothesis Statement :

Null Hypothesis : The mean cost of Gasoline Trucks and Diesel Trucks are same .

$$H_{o\,:}\,\mu_G = \mu_D$$

Alternate Hypothesis : The mean cost of Gasoline Trucks and Diesel Trucks are significantly difference .

$$H_{1\,:}\,\mu_G \neq \mu_D$$

Step 2 :- Data Analysis :

```
library(readxl)
data<-read_excel("D:/MULTIVARIATE WORK/lab 8.xlsx")

## New names:
## • `X1` -> `X1...1`
## • `X2` -> `X2...2`
## • `X3` -> `X3...3`
## • `X1` -> `X1...4`
## • `X2` -> `X2...5`
## • `X3` -> `X3...6`

first_matrix<-matrix(data=c(data$X1...1,data$X2...2,data$X3...3),ncol = 3 )

second_matrix<-matrix(data=c(data$X1...4,data$X2...5,data$X3...6),ncol=3)

diffrence<-first_matrix-second_matrix

y<-colMeans(diffrence)

covarince<-cov(diffrence)

inverse<-solve(covarince)

t_sq<-15%*%t(y)%*%inverse%*%(y)

f_cal<-((15-3)*t_sq)/(3*(15-1))
f_cal

##          [,1]
## [1,] 7.799979
```

Step 3 :- Conclusion :

So, at the $\alpha = 1\%$ level of Significance

F_cal = 7.79 > 5.56, by according to the given data we rejected the Null Hypothesis .

Step 4 :- Interpretation :

The mean cost of Gasoline Trucks and Diesel Trucks are significantly differ .